

FTC Autonomous Student Workbook

Part 9 — Drive Forward Three Feet and Stop

Name _____ Team _____ Date _____

You will turn 36 inches into encoder counts, trace an autonomous program, conduct safe measured trials, and use evidence to calibrate the robot.

Learning targets

- ☐ I can explain autonomous versus TeleOp.
- ☐ I can calculate counts per inch and a target count.
- ☐ I can trace RUN_TO_POSITION and every stop path.
- ☐ I can perform POWERING ON correctly.
- ☐ I can measure repeated trials and propose a calibration update.

Before the video

How could a robot estimate three feet without a joystick? What could make the estimate wrong?

Video focus notes

TIME	FOCUS QUESTION	MY NOTES
0:09	What replaces the joystick in autonomous?	
0:18	What is our complete first mission?	
0:29	Why must one drive motor be reversed?	
0:39	What does an encoder measure—and what does it not measure?	
0:49	How does wheel circumference connect counts to inches?	

TIME	FOCUS QUESTION	MY NOTES
0:58	Why is the target about 1,630 rather than exactly 1,630?	
1:09	What happens between POWERING ON and START?	
1:29	What three commands begin the move?	
1:39	Why do we need STOP and a timeout in addition to encoder targets?	
1:49	Does BRAKE make an enabled robot safe to approach?	
2:00	Why measure several trials?	

Hardware assumption audit

ITEM	EXAMPLE VALUE	OUR ROBOT	PASS
Drive motor	19.2:1 Yellow Jacket	537.7 counts/rev	
Drive wheel	96 mm	3.7795 in	
Motor-to-wheel reduction	1.0	direct 1:1	
Hardware names	left_drive / right_drive	exact match	
Encoder cables	both connected	counts change	

Calculate the target

1. $96 \text{ mm} / 25.4 = \underline{\hspace{2cm}}$ inches
2. Wheel circumference = diameter x pi = $\underline{\hspace{2cm}}$ inches
3. Counts per inch = $537.7 / \text{circumference} = \underline{\hspace{2cm}}$
4. $36 \text{ inches} \times \text{counts per inch} = \underline{\hspace{2cm}}$ counts
5. Rounded whole-count target = $\underline{\hspace{2cm}}$ counts

Trace the code

Number these from 1 through 9.

- ☐ set both target positions
- ☐ waitForStart

- ☐ command zero motor power
- ☐ map left and right motors
- ☐ monitor isBusy and timeout
- ☐ reset encoders
- ☐ select RUN_TO_POSITION
- ☐ apply power 0.35
- ☐ set directions and BRAKE

Stop-path map

Complete: Normal finish, Driver Station _____, or six-second _____ all lead to motor power _____.

POWERING ON trial checklist

STEP	ACTION	INITIALS
1	Robot square at start while DISABLED	
2	Hands, tools, tape measure, and cables removed	
3	Full lane, side boundary, and stopping buffer checked	
4	POWERING ON called loudly	
5	Everyone given time to step behind boundary	
6	Safety captain visually confirmed clear	
7	Operator ready at STOP before START	

Measured trial record

Do not enter the lane until STOP is pressed and DISABLED is visually confirmed. Measure from the same robot reference point every time.

TRIAL	COMMAND IN	ACTUAL IN	LEFT FINAL	RIGHT FINAL	OBSERVATION
1	36.0				

TRIAL	COMMAND IN	ACTUAL IN	LEFT FINAL	RIGHT FINAL	OBSERVATION
2	36.0				
3	36.0				
4	36.0				
5	36.0				

Average actual distance = _____ inches

Did the trials agree closely? YES / NO Evidence: _____

Calibration

new counts/inch = old counts/inch x commanded distance / actual distance

Old CPI _____ x commanded _____ / average actual _____ = new CPI _____

If the robot traveled short, the new CPI should: INCREASE / DECREASE

Confirmation trial actual distance: _____ inches Error from 36: _____ inches

Evidence-based troubleshooting

EVIDENCE	QUESTION	MY RECORD
Code	Which constant, target, mode, power, and stop path ran?	
Telemetry	What were target and final left/right counts?	
Measurement	What was the actual distance from the reference point?	
Mechanics	Was travel straight and repeatable? Was slip visible?	
Hypothesis	Which single cause best fits all evidence?	
Next safe test	What smallest test could confirm it?	

Exit response

Explain how 36 inches becomes a motor target, why the real distance may differ, and how the program and team both make the test stop safely.

I understand that stopped motion is not permission to approach. I must press STOP, confirm DISABLED, and receive permission first.

Student initials _____ Safety captain initials _____